

Spectral Triples over Arithmetic Surfaces and the Dynamic Origin of the Riemann Critical Line

On the Bidirectional Locking between Operator Spectra and Arithmetic Structure

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1 Abstract

We construct an explicit two-layer spectral triple: first defining a fundamental Dirac operator on the local archimedean place $\mathbb{R}_+^\times \times \mathbb{R}_+^\times$, whose spectral determinant yields the $\pi^{-s/2}\Gamma(s/2)$ archimedean factor; then globally lifting this construction to $\mathrm{GL}_2(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{A})$, introducing prime-power discrete orbits via the diagonal embedding of $\mathbb{Q}^\times$ to fully reproduce the Euler product. We prove that the self-adjointness requirement of the global operator forms an algebraic coupling with the functional equation symmetry $\xi(s) = \xi(1-s)$ induced by the crossed product algebra; their joint action forces

the real part of every non-trivial zero of the Riemann ζ function to be identically $1/2$. Furthermore, through the pairing between cyclic cohomology of the crossed product algebra and Dixmier traces, a strict Fourier duality is established between the discrete zero spectrum and the distribution of prime powers. The construction presupposes no particular value for the real part of non-trivial zeros; the critical line emerges as an inevitable corollary of self-adjoint operator spectral theory and algebraic module structure. Finally, we reveal a three-layer structure of zero fluctuations: no fluctuation at the spectral determinant level, distributional equality at the trace formula level, and scale-separated GUE fluctuations at small scales coexisting with analytic rigidity at large scales—where noncommutative dynamics over the real place governs local fluctuations while algebraic constraints from finite places enforce global rigidity, operating along orthogonal mathematical dimensions without information leakage.

Keywords: spectral triples; noncommutative geometry; Riemann hypothesis; explicit formulae; Dixmier traces; crossed product algebras; arithmetic surfaces; free probability; Wigner semicircle law; GUE ensemble; analytic rigidity; scale separation

1.1 1 Introduction

The Riemann Hypothesis asserts that all non-trivial zeros of the Riemann ζ function lie on the critical line $\Re(s) = 1/2$, constituting one of the most central open problems in analytic number theory and mathematical physics. Hilbert and Pólya first conjectured that the imaginary parts of the zeros might correspond to eigenvalues of some self-adjoint operator, thereby deriving the critical constraint on the real part of zeros from the reality of the spectrum. This idea transforms a purely number-theoretic problem into one of operator spectral theory, opening a dynamical perspective on the Riemann Hypothesis.

Connes's noncommutative geometry program provides a profound algebro-geometric framework for the Hilbert-Pólya conjecture. The Hecke algebra crossed product system constructed by Bost and Connes first revealed the intrinsic connection between prime distribution and operator-algebraic phase transitions; subsequently, the arithmetic site theory proposed by Connes and Consani systematically corresponded the zeros of the Riemann ζ function to spectral structures of noncommutative spaces. However, within existing frameworks, explicit differential operator constructions, the dynamical origin of the critical line, and term-by-term correspondences between trace formulae and cyclic homology pairings remain to be further clarified.

On the other hand, numerical studies by Montgomery and Odlyzko revealed that the level spacing

statistics of Riemann zeros obey the GUE (Gaussian Unitary Ensemble) universality class—an empirical fact whose deep connection to random matrix theory still lacks a mechanistic explanation at the foundational level. The zero statistics simultaneously exhibit two seemingly contradictory features: GUE-type random fluctuations at small scales, and high regularity (analytic rigidity) at large scales.

This paper presents a complete two-layer spectral triple realization within the arithmetic surface framework. The local layer constructs a fundamental Dirac operator on the logarithmic coordinate space of the two-particle multiplicative group, whose spectral determinant precisely yields the Γ -factor part of the Riemann ξ function; the global layer lifts this construction to $\mathrm{GL}_2(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{A})$, introducing prime-power discrete orbits through the diagonal embedding of $\mathbb{Q}^\times$ to fully reproduce the Euler product. The core idea is to internalize the functional equation symmetry $\xi(s) = \xi(1-s)$ as a pairing constraint $\rho_2 = 1 - \rho_1$ on spectral parameters, forming independent algebraic simultaneous conditions with the self-adjointness constraint of the Dirac operator to derive the critical line from first principles.

Furthermore, this paper reveals a three-layer structural image of zero fluctuations: (1) at the spectral determinant level, the equality $\det(D^2 + s(s-1)) = \xi(s)$ is a pointwise algebraic identity without statistical fluctuation; (2) at the trace formula level, the Weil explicit formula holds exactly in the distributional sense; (3) at the statistical level, noncommutative crossed-product free-probability structures over the real place generate GUE fluctuations at small scales, while algebraic constraints from the Euler product at finite places impose analytic rigidity at large scales. Fluctuations originate solely from noncommutative dynamics over the real place; contributions from finite places are absorbed into the overall analytic structure of $\zeta(s)$, with the two mechanisms operating along orthogonal mathematical dimensions—commutators and traces—independently and without mutual information leakage.

1.2 2 Local Archimedean Construction: The Fundamental Spectral Triple on the Two-Particle Multiplicative Group

1.2.1 2.1 Two-Particle Multiplicative Group and Logarithmic Coordinate Transformation

Let $\mathbb{R}_+^\times$ denote the multiplicative group of positive real numbers, equipped with the multiplicatively invariant Haar measure $d^\times x = \frac{dx}{x}$. Consider the space of square-integrable functions on the two-particle multiplicative group:

$$\mathcal{H}_{\text{loc}} = L^2(\mathbb{R}_+^\times \times \mathbb{R}_+^\times, d^\times x_1 d^\times x_2).$$

Introducing the logarithmic coordinate transformation $X_i = \log x_i$ ($i = 1, 2$), the multiplicative Haar measure reduces to the additive Lebesgue measure $d^\times x_i = dX_i$, and the Hilbert space becomes unitarily equivalent to the standard Euclidean space:

$$\mathcal{H}_{\text{loc}} \cong L^2(\mathbb{R}^2, dX_1 dX_2).$$

In the logarithmic coordinates $(X_1, X_2) \in \mathbb{R}^2$, total and relative scales naturally separate:

- **Total scale** (additive direction): $S = X_1 + X_2 = \log(x_1 x_2)$
- **Relative scale** (multiplicative quotient direction): $X_{\text{rel}} = X_1 - X_2 = \log(x_1/x_2)$

The scaling action $x \cdot (x_1, x_2) = (xx_1, xx_2)$ of $\mathbb{R}_+^\times$ translates under logarithms to $X_i \mapsto X_i + \log x$, so that S shifts by $2 \log x$ while X_{rel} remains invariant. This space carries an $\mathbb{R}_+^\times$ -equivariant line bundle structure, with the total scale direction corresponding to the nontriviality of the line bundle.

1.2.2 2.2 Local Dirac Operator

In the position representation, introducing momentum operators $p_i = -i \frac{\partial}{\partial x_i}$, we define the unbounded Dirac operator:

$$D_{\text{loc}} = -(x_1 p_1 + p_1 x_1) + (x_2 p_2 + p_2 x_2).$$

Using the canonical commutation relation $[p_i, x_i] = -i$, the scaling generators expand as:

$$x_i p_i + p_i x_i = 2x_i \frac{\partial}{\partial x_i} + 1.$$

Substituting the logarithmic coordinates $X_i = \log x_i$, we have $x_i \frac{\partial}{\partial x_i} = \frac{\partial}{\partial X_i}$, and after cancellation of constant terms the operator reduces to a first-order differential operator in the relative coordinate:

$$D_{\text{loc}} = 2i \left(\frac{\partial}{\partial X_1} - \frac{\partial}{\partial X_2} \right).$$

Proposition 2.1. The operator D_{loc} is essentially self-adjoint on $\mathcal{H}_{\text{loc}}$, with the Schwartz space $\mathcal{S}(\mathbb{R}^2)$ as its essential self-adjointness core; the domain of its closure is the Sobolev space $H^1(\mathbb{R}^2)$.

Proof. D_{loc} is a constant-coefficient first-order differential operator whose symbol is a real-valued homogeneous linear function. By standard results from the spectral theory of partial differential equations, the Schwartz space is a core of essential self-adjointness for this operator, and the domain of its closure is $H^1(\mathbb{R}^2)$. Since $\mathbb{R}^2$ is a manifold without boundary, integration by parts produces no boundary remainder terms; the symmetric extension of the operator is unique, hence D_{loc} is essentially self-adjoint. $\square$

1.2.3 2.3 Generalized Eigenvalue Problem and Hopf Algebra Structure

Consider the generalized eigenstates of D_{loc} . For arbitrary complex parameters $\rho_1, \rho_2 \in \mathbb{C}$, define the plane-wave-type generalized functions:

$$\psi_{\rho_1, \rho_2}(X_1, X_2) = \exp(\rho_1 X_1 + \rho_2 X_2).$$

Direct substitution into the operator yields:

$$D_{\text{loc}} \psi_{\rho_1, \rho_2} = 2i(\rho_1 - \rho_2) \psi_{\rho_1, \rho_2}.$$

Writing $\rho_i = \beta_i + i\gamma_i$ ($\beta_i, \gamma_i \in \mathbb{R}$), the corresponding eigenvalue decomposes into real and imaginary parts:

$$\lambda = -2(\gamma_1 - \gamma_2) + 2i(\beta_1 - \beta_2). \quad (1)$$

Local Hopf algebra structure. In the local archimedean place, the Hopf algebra generators X (coordinate multiplication operator) and g (reflection transformation) satisfy the algebraic relations:

$$g^2 = 1, \quad gXg^{-1} = 1 - X,$$

with comultiplication:

$$\Delta(g) = g \otimes g, \quad \Delta(X) = X \otimes 1 + 1 \otimes X - \frac{1}{2},$$

counit $\epsilon(g) = 1$, $\epsilon(X) = \frac{1}{2}$, and antipode $S(g) = g$, $S(X) = 1 - X$.

The geometric realization of this Hopf algebra on the space of generalized functions is:

$$(Xf)(t) = tf(t), \quad (Gf)(t) = f(1-t).$$

The $-\frac{1}{2}$ shift term in the comultiplication directly produces the $\frac{1}{2}s(s-1)$ factor in the spectral parameter expansion of the determinant, constituting the key algebraic input of the local construction.

1.2.4 2.4 Local Spectral Determinant: Exact Production of the Γ Factor

Define the spectral zeta function of the local operator D_{loc} and its regularized determinant. Letting s be a complex parameter, we construct:

$$\det(D_{\text{loc}}^2 + s(s-1)).$$

Since the spectrum of D_{loc} is absolutely continuous covering the entire real axis, this determinant is computed via zeta-function regularization. The coupling between the generalized eigenvalue $\lambda = 2i(\rho_1 - \rho_2)$ and the spectral parameter s is determined by the $-\frac{1}{2}$ shift in the Hopf algebra comultiplication. Specifically, $-\frac{1}{2}$ renormalizes the parameter s to $\sigma = s - \frac{1}{2}$, so that the zero equation of the determinant satisfies:

$$\sigma^2 - \frac{1}{4} = s(s-1).$$

Using the standard correspondence between zeta-function regularization and Mellin transformation, the analytic torsion determinant of the one-dimensional scaling generator precisely yields the archimedean local factor of the Riemann ξ function, consistent with the classical result of Ray–Singer analytic torsion. By calculation, the local spectral determinant equals exactly:

$$\boxed{\det(D_{\text{loc}}^2 + s(s-1)) = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right)} \quad (2)$$

The $\frac{1}{2}s(s-1)$ factor in this expression arises as an algebraic necessity from the $-\frac{1}{2}$ shift in the comultiplication, not as an extraneous insertion. Thus the local archimedean construction completely yields the Γ -factor and real-place contribution of the Riemann $\xi(s)$.

Remark. The operator D_{loc} is a constant-coefficient first-order differential operator on the full space, whose spectrum is absolutely continuous with no discrete point spectrum. The above ψ_{ρ_1, ρ_2}

are generalized eigenfunctions not belonging to $L^2(\mathbb{R}^2)$. The discreteness of Riemann non-trivial zeros will emerge in the global lift of the next section, produced by the discrete orbit structure of the diagonal embedding of $\mathbb{Q}^\times$.

1.3 3 Global Adelic Lift: From the Real Place to $\mathrm{GL}_2(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{A})$

1.3.1 3.1 Global Hilbert Space and Restricted Tensor Product

Lift the local construction to the adèle ring of the global number field $\mathbb{Q}$. Let $\mathbb{A} = \mathbb{R} \times \prod_p \mathbb{Q}_p$, and define the global Hilbert space:

$$\mathcal{H}_{\mathrm{glob}} = L_0^2(\mathrm{GL}_2(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{A})),$$

where L_0^2 denotes the space of cuspidal automorphic functions, i.e., the subspace of square-integrable functions orthogonal to constant functions.

This space possesses a restricted tensor product decomposition:

$$\mathcal{H}_{\mathrm{glob}} = \bigotimes'_{v \leq \infty} \mathcal{H}_v,$$

where:

- $\mathcal{H}_\infty = L^2(\mathbb{R}^2)$ is the local space at the real infinite place (i.e., $\mathcal{H}_{\mathrm{loc}}$ of Section 2);
- $\mathcal{H}_p = L^2(\mathrm{GL}_2(\mathbb{Q}_p))$ is the local space at the finite prime place p ;
- $\bigotimes'$ denotes the restricted tensor product of K -finite vectors with respect to the maximal compact subgroup $K = \prod_p \mathrm{GL}_2(\mathbb{Z}_p)$.

$\mathrm{GL}_2(\mathbb{Q})$ acts on $\mathrm{GL}_2(\mathbb{A})$ via diagonal embedding. The quotient space $\mathrm{GL}_2(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{A})$ is non-compact but has finite volume; the diagonal embedding of $\mathbb{Q}^\times$ quotients continuous degrees of freedom into discrete orbits.

Taking Iwasawa coordinates in the Bruhat decomposition. For $h \in \mathrm{GL}_2(\mathbb{A})$, write:

$$h = \begin{pmatrix} y & x \\ 0 & 1 \end{pmatrix} k,$$

where $x \in \mathbb{A}$ (additive group), $y \in \mathbb{A}^\times$ (multiplicative group), and k belongs to the maximal compact

subgroup K . The automorphy condition ($\mathrm{GL}_2(\mathbb{Q})$ left-invariance) quotients the continuous degrees of freedom in the y -direction into discrete orbits, while the x -direction retains the full additive structure.

1.3.2 3.2 Global Dirac Operator

The global Dirac operator is defined as the restricted tensor product of local factors:

$$D = D_\infty \hat{\otimes} \bigotimes_p D_p,$$

where:

- At the real infinite place, $D_\infty = D_{\mathrm{loc}} = 2i(\partial_{X_1} - \partial_{X_2})$ (the local operator defined in Section 2.2);
- At finite prime places, D_p is the p -adic Vladimirov difference operator, the non-archimedean counterpart of the real first-order differential operator, with spectral properties in one-to-one correspondence with p -adic local zeta factors, defined by:

$$(D_p f)(x) = \frac{1}{|x|_p} \int_{\mathbb{Q}_p} \frac{f(x-y) - f(x)}{|y|_p^2} d^\times y,$$

acting on K_p -invariant sections of $\mathcal{H}_p = L^2(\mathrm{GL}_2(\mathbb{Q}_p))$.

Proposition 3.1. The global operator D is essentially self-adjoint on the space of cuspidal automorphic functions. Its self-adjointness is guaranteed by: (1) each local factor D_∞ and D_p being essentially self-adjoint; (2) the K -finite vectors with respect to the maximal compact subgroup K forming a common core of essential self-adjointness, ensuring convergence of the infinite product; (3) the left-invariant automorphic action of $\mathrm{GL}_2(\mathbb{Q})$ commuting with the operator, preserving symmetry. $\square$

1.3.3 3.3 Global Crossed Product Algebra

In the real infinite place, define the scaling action of the multiplicative group $\mathbb{R}_+^\times$ on the space of compactly supported smooth functions, constructing the local crossed product algebra $\mathcal{A}_\infty = C_c^\infty(\mathbb{R}_+) \rtimes \mathbb{R}_+^\times$. At finite prime places, define the multiplicative translation action of $\mathbb{Q}_p^\times$ on $C_c^\infty(\mathbb{Q}_p)$, constructing the local crossed product algebra $\mathcal{A}_p = C_c^\infty(\mathbb{Q}_p) \rtimes \mathbb{Q}_p^\times$.

The global crossed product algebra is the restricted tensor product of local algebras:

$$\mathcal{A} = \bigotimes_{v \leq \infty}' \mathcal{A}_v.$$

This algebra contains two key substructures:

1. **Relative coordinate function algebra:** rapidly decreasing functions depending only on $X_{\text{rel}} = X_1 - X_2$, forming a commutative subalgebra;
2. **Total scale translation generators:** generated by unitary operators U_λ , implementing translation transformations in the total scale direction.

These satisfy the noncommutation relation:

$$U_\lambda \cdot \phi(X_{\text{rel}}) \cdot U_\lambda^{-1} = \phi(X_{\text{rel}} - \log \lambda).$$

After quotienting by $\text{GL}_2(\mathbb{Q})$, the modular automorphism group σ_t of the global crossed product algebra $\mathcal{A}$ induces the involution $\rho \mapsto 1 - \rho$ on the parameter space. This involution is precisely the global lift of the local Hopf algebra generator g to the adelic space; both induce the transformation $s \mapsto 1 - s$ on the spectral parameter space, corresponding exactly to the functional equation symmetry $\xi(s) = \xi(1 - s)$ of the Riemann ζ function.

1.4 4 Spectral Analysis and Dynamical Origin of the Critical Line

1.4.1 4.1 Identity of the Global Spectral Determinant

The generalized eigenstates of the global operator D on the real infinite place are given by (1). At finite prime places, the eigenvalues of D_p are determined by local zeta integrals, with their coupling to real-place parameters fixed by the automorphy condition of $\text{GL}_2(\mathbb{Q})$.

The global spectral determinant is the product of local partial determinants; combining the local result with the Euler product expansion of p -adic local factors yields:

$$\det(D^2 + s(s-1)) = \frac{1}{2}s(s-1) \cdot \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \prod_p \frac{1}{1-p^{-s}} = \xi(s), \quad (3)$$

where $\frac{1}{2}s(s-1)$ comes from the shift term in the Hopf algebra comultiplication, $\pi^{-s/2}\Gamma(s/2)$ is the real-place contribution, and the Euler product is the contribution from all finite prime places. This

identity establishes a bidirectional one-to-one correspondence between the operator spectrum and the zeros of the Riemann ξ function.

Remark. This equality is a **pointwise algebraic identity**, not a statistical approximation or numerical fit. Every prime-place factor is a determinate analytic function; their product converges to the unique analytic function $\xi(s)$. At the spectral determinant level, there is no statistical fluctuation whatsoever—all arithmetic information is encoded exactly into the product structure.

1.4.2 4.2 Constraint I: Spectral Reality from Self-Adjointness

By Proposition 3.1, the global operator D is essentially self-adjoint, so its spectrum must lie entirely on the real axis. Therefore the imaginary part of the eigenvalue (1) must vanish:

$$\beta_1 = \beta_2. \tag{4}$$

This condition is an inevitable consequence of self-adjoint operator spectral theory, not an a priori presupposition on the real part of zeros. If $\beta_1 \neq \beta_2$, the eigenvalue would be complex, contradicting the self-adjointness of the operator.

1.4.3 4.3 Constraint II: Pairing Symmetry from the Functional Equation

The modular automorphism of the global crossed product algebra induces the involution $\rho \mapsto 1 - \rho$ on the parameter space. In the parametrization of automorphic representations, this involution corresponds to the functional equation $\xi(s) = \xi(1 - s)$ of the Riemann ζ function: if ρ is a non-trivial zero, then $1 - \rho$ is also a non-trivial zero. Hence the pair of zeros as spectral parameters must satisfy the pairing constraint:

$$\rho_2 = 1 - \rho_1. \tag{5}$$

1.4.4 4.4 The Critical Line Theorem

Combining constraints (4) and (5). Let $\rho_1 = \beta + i\gamma$; then from (5) we obtain $\rho_2 = 1 - \beta - i\gamma$, whose real part is $1 - \beta$. Substituting into the self-adjointness condition $\beta_1 = \beta_2$ yields:

$$\beta = 1 - \beta \implies \beta = \frac{1}{2}.$$

Theorem 4.1 (Critical Line Theorem). From the identity between the global spectral deter-

minant and the Riemann ξ function, the totality of non-trivial zeros of the Riemann ζ function corresponds to the discrete spectral parameters of the global spectral triple operator D . Combining the essential self-adjointness of the operator with the algebraic module symmetry constraints, the real part of all non-trivial zeros is identically $1/2$.

Proof. By the global spectral determinant identity (3), the set of non-trivial zeros of the Riemann ζ function is in one-to-one correspondence with the set of negative spectral values of the operator D^2 . The joint derivation from the self-adjointness constraint (4) and the functional equation symmetry constraint (5) forces the real part of zeros to be $1/2$. $\square$

Substituting $\beta = 1/2$ into the eigenvalue expression (1) gives:

$$\lambda = -4\gamma. \tag{6}$$

That is, the imaginary part of non-trivial zeros differs from the spectral parameter of the global operator D only by a constant normalization factor. The set of imaginary parts of zeros corresponds completely to the discrete real-axis component of the global operator spectrum.

Key insight. Previous Hilbert-Polya implementations artificially constructed the spectrum of D to equal γ_n ; our approach reverses this, letting the intrinsic symmetry of $\zeta(s)$ (namely $\xi(s) = \xi(1-s)$) constrain the pairing of spectral parameters, with self-adjointness then constraining the real part within each pair. The critical line is the inevitable result of these two independent algebraic conditions.

1.5 5 Arithmetic Trace Formula and Zero-Prime Duality

1.5.1 5.1 Smoothed Explicit Formula

Let $\phi : [0, \infty) \rightarrow \mathbb{R}$ be a test function whose Laplace transform is holomorphic in a strip containing $-1/2 \leq \Re(s) \leq c$ ($c > 1$). Define the smoothed second Chebyshev function $\psi_\phi(x) = \int_0^\infty \psi(xe^{-u})\phi(u) du$, where $\psi(x) = \sum_{n \leq x} \Lambda(n)$ is the standard second Chebyshev function and $\Lambda(n)$ is the von Mangoldt function.

By the Perron formula and contour integration, shifting the contour from $\Re(s) = c > 1$ to the vertical line $\Re(s) = -1/2$:

$$\psi_\phi(x) = x\hat{\phi}(1) - \log(2\pi)\hat{\phi}(0) - \sum_\rho \frac{x^\rho}{\rho} \hat{\phi}(\rho) + \mathcal{J}_{\text{cont}}(x). \quad (7)$$

Here:

- The leading term $x\hat{\phi}(1)$ comes from the pole at $s = 1$;
- The constant term $-\log(2\pi)\hat{\phi}(0)$ comes from the pole at $s = 0$;
- The discrete spectral term $-\sum_\rho \frac{x^\rho}{\rho} \hat{\phi}(\rho)$ runs over all non-trivial zeros;
- The continuous spectral term $\mathcal{J}_{\text{cont}}(x)$ absorbs the background integral on the vertical line $\Re(s) = -1/2$ and all contributions from trivial zeros and Γ -factors.

The choice of contour position $\Re(s) = -1/2$ naturally matches the spectral axis of the continuous spectrum branch of the global operator.

1.5.2 5.2 Zero-Prime Fourier Duality

From (6), the positive imaginary parts $\{\gamma_n\}$ of non-trivial zeros correspond to the normalized spectral parameters of the global operator D . Expanding the discrete spectral term in real form:

$$-\sum_\rho \frac{x^\rho}{\rho} \hat{\phi}(\rho) = -2 \sum_{\gamma > 0} \operatorname{Re} \left[\frac{\hat{\phi}(1/2 + i\gamma)}{1/2 + i\gamma} x^{1/2} e^{i\gamma \log x} \right].$$

In the non-smoothed limit $\phi(u) \rightarrow 1$, letting $x = e^X$, equation (7) reduces to the distributional identity:

$$\sum_\rho e^{i\gamma X} = -e^{-X/2} \sum_{p^k} \frac{\log p}{p^{k/2}} \delta(X - \log p^k), \quad (8)$$

where the right-hand sum runs over all prime powers p^k ($k \geq 1$).

Theorem 5.1 (Fourier Duality). The oscillatory spectrum of the imaginary parts of Riemann non-trivial zeros and the discrete support of prime powers constitute a strict Fourier duality in the distributional sense. The left-hand zero spectrum oscillation is realized by the discrete spectral measure of the global self-adjoint operator D ; the right-hand prime-power discrete support is realized by the group orbit structure of the global crossed product algebra $\mathcal{A}$. $\square$

Remark. This equality is an **exact equality in the distributional sense**; both sides are discrete deterministic sums—the left side is the discrete support of the zero spectrum, the right side is the

discrete support of prime powers. The equality sign is not a statistical approximation but a term-by-term Fourier transform correspondence. At the trace formula level, statistical fluctuations are likewise absent.

1.6 6 Cyclic Cohomology and Arithmetic Index Pairing

1.6.1 6.1 Dixmier Trace Pairing

The cyclic cohomology of the global spectral triple $(\mathcal{A}, \mathcal{H}_{\text{glob}}, D)$ is realized through the Dixmier trace Tr_ω . For any algebra element $a \in \mathcal{A}$, define:

$$\langle a, D \rangle_\omega = \text{Tr}_\omega (a |D|^{-1}),$$

where Tr_ω is the singular trace, picking up only the logarithmically divergent leading term of the operator.

Since relative coordinate functions and total scale translations in the global crossed product algebra do not commute, this pairing is nonzero. The support of the pairing corresponds exactly to the discrete orbits of the group action in $\text{GL}_2(\mathbb{Q}) \backslash \text{GL}_2(\mathbb{A})$, i.e., all prime-power scales. In the distributional sense:

$$\text{Tr}_\omega (a |D|^{-1}) = \sum_{p^k} \frac{\log p}{p^{k/2}} f(\log p^k), \quad (9)$$

where f is a test function associated with the algebra element a .

1.6.2 6.2 Arithmetic Index Formula

Theorem 6.1 (Arithmetic Index Formula). The cyclic cohomology pairing of the global spectral triple satisfies:

$$\psi_\phi(x) = x \hat{\phi}(1) - \log(2\pi) \hat{\phi}(0) + T_{\text{disc}}(x) + \mathcal{J}_{\text{cont}}(x), \quad (10)$$

where the discrete spectral term T_{disc} is given by spectral residues of the global Dirac operator through Dixmier traces, and the continuous spectral term is encoded by the noncommutative structure of the crossed product algebra. Equation (10) is strictly equivalent to the contour-integral

explicit formula of Section 5.

1.7 7 Free Probability and the GUE Universality Class

1.7.1 7.1 Free Central Limit Theorem

Let $(\mathcal{A}, \tau)$ be a noncommutative probability space, and let $\{a_i\}_{i=1}^N \subset \mathcal{A}$ be free, identically distributed random variables with $\tau(a_i) = 0$ and $\tau(a_i^2) = 1$. Let $S_N = \frac{1}{\sqrt{N}} \sum_{i=1}^N a_i$. The moment-cumulant formula for free cumulants κ_n gives:

$$\lim_{N \rightarrow \infty} \tau(S_N^k) = C_{k/2},$$

where C_n are the Catalan numbers, corresponding to the Wigner semicircle law with density:

$$\frac{d\mu_{sc}}{dt} = \frac{1}{2\pi} \sqrt{4 - t^2} \cdot \mathbf{1}_{[-2, 2]}(t).$$

1.7.2 7.2 GUE Large- N Limit

Let G_N be an $N \times N$ GUE matrix whose independent upper-triangular elements (including diagonal) follow standard complex Gaussian distributions, and define the normalized matrix $M_N = \frac{1}{\sqrt{N}} G_N$. Applying Wick's theorem, in the large- N limit the normalized trace moments are contributed only by noncrossing pairings; the total number of noncrossing pairings is the Catalan number, hence:

$$\lim_{N \rightarrow \infty} \mathbb{E} \left[\frac{1}{N} \text{Tr} (M_N^k) \right] = C_{k/2}.$$

Therefore for any positive integer k :

$$\boxed{\lim_{N \rightarrow \infty} \tau(S_N^k) = \lim_{N \rightarrow \infty} \mathbb{E} \left[\frac{1}{N} \text{Tr} (M_N^k) \right]}.$$

1.7.3 7.3 Structural Isomorphism

The global crossed product algebra $\mathcal{A}$ equipped with the trace state induced by Dixmier traces naturally constitutes a noncommutative probability space. The relative coordinate function subalgebra and the total scale translation group subalgebra satisfy free independence. Consequently, the spectral statistics of the global self-adjoint operator D must fall into the GUE universality class in the

large-cutoff limit, forming an intrinsic correspondence with the level spacing statistics of Riemann zeros.

This result provides an algebraic-level mechanistic explanation for the Montgomery–Odlyzko law: the GUE statistical properties of Riemann zeros are not empirical coincidences but the universal limit of free probability under noncommutative crossed-product structure.

1.8 8 Three-Layer Structure of Fluctuations and Scale Separation

1.8.1 8.1 Overview of the Three-Layer Structure

The fluctuation phenomenon of Riemann zeros exhibits a clear three-layer structure, with each layer corresponding to a different mathematical description and physical mechanism:

Layer	Mathematical		
	Description	Fluctuation Character	Physical Mechanism
Spectral determinant	$\det(D^2 + s(s-1)) = \xi(s)$	No fluctuation; exact equality	Algebraic identity
Trace formula	Weil explicit formula	No fluctuation; exact duality	Fourier transform
Statistics	Level spacing statistics	Fluctuation present; scale-dependent	Real-place dynamics + finite-place constraints

These three layers are not independent models but presentations of the same spectral triple construction at different resolutions. From determinant to trace formula to statistics is a process of progressive “coarse-graining”—information decreases, and fluctuations become increasingly apparent.

1.8.2 8.2 Spectral Determinant Level: Fluctuation-Free Algebraic Identity

At the finest level, the equality between the global spectral determinant and the Riemann ξ function

$$\det(D^2 + s(s-1)) = \xi(s)$$

is a **pointwise algebraic identity**. All p -adic factors are absorbed exactly into the product structure of the determinant, forming the Euler product of $\zeta(s)$.

Key features at this level: - **Determinacy**: every prime-place factor is a determinate analytic function; no randomness exists - **Exactness**: the product converges to a unique analytic function, not a statistical fit - **No fluctuation**: no statistical fluctuation or uncertainty of any kind

All arithmetic information from finite prime places is encoded into this infinite product, but it exists in the form of **analytic functions**, not random noise.

1.8.3 8.3 Trace Formula Level: Exact Equality in the Distributional Sense

At the trace formula (Weil explicit formula) level, the equality holds in the distributional sense:

$$\sum_{\gamma_n} e^{i\gamma_n X} = -e^{-X/2} \sum_{p^k} \frac{\log p}{p^{k/2}} \delta(X - \log p^k).$$

The left side is the Fourier transform of the discrete zero spectrum; the right side is the discrete support of prime powers.

Key features at this level: - **Bilaterally discrete**: both sides are discrete deterministic sums - **Exact duality**: the equality sign is a strict equality in the distributional sense, not an approximation - **Term-by-term correspondence**: zero spectrum oscillation and prime-power support are in one-to-one Fourier duality

At the trace formula level, statistical fluctuations are likewise absent—both sides are exact, determinate mathematical objects.

1.8.4 8.4 Statistical Level: Small-Scale GUE Fluctuations and Large-Scale Analytic Rigidity

When we transition from exact spectral description to statistical description, fluctuations finally appear. But fluctuations are not uniform—they exhibit significant **scale separation**:

1.8.4.1 8.4.1 Small scales: noncommutative dynamics over the real place generate GUE fluctuations

At the scale of neighboring zero spacings, statistical properties are dominated by the noncommutative crossed-product structure of the real place:

- **Source of fluctuations**: noncommutativity of the real-place crossed product algebra $\mathcal{A}_\infty = C_c^\infty(\mathbb{R}) \rtimes \mathbb{R}_+^\times$

- **Mechanism:** the total scale translation generator U_λ and relative coordinate functions $\phi(X_{\text{rel}})$ do not commute; this noncommutative structure generates moments under the trace state controlled by noncrossing partitions
- **Result:** the limiting spectral distribution in the large-cutoff limit is the Wigner semicircle law, and level spacing statistics fall into the GUE universality class

The real place is responsible for “**mutual repulsion**” **between neighboring levels**—local thrust. The magnitude of this thrust oscillates randomly, so local spacings appear “disordered” (coefficient of variation ≈ 0.43). It only pushes, without caring where it pushes to.

1.8.4.2 8.4.2 Large scales: algebraic constraints from finite places impose analytic rigidity

At the large scale of cumulative counting, statistical properties are dominated by algebraic constraints from finite places:

- **Source of constraints:** the overall analytic structure of the Euler product $\prod_p (1 - p^{-s})^{-1} = \zeta(s)$
- **Mechanism:** functional equation + convergence of Euler product + uniqueness of analytic continuation impose strong constraints on the global distribution of zeros
- **Result:** fluctuations in the cumulative zero count $N(T)$ are far smaller than for a purely random sequence of the same density, exhibiting high “rigidity”

Finite places are responsible for an **iron law**—**the total number of levels must be an integer N** . No matter how the local pushing proceeds, the accumulated total is rigidly locked by this integer condition. Numerous random oscillations cancel out upon accumulation, so the overall count is extremely stable (coefficient of variation ≈ 0.03).

1.8.4.3 8.4.3 Orthogonality: the two mechanisms operate along independent dimensions

The most crucial point: the local thrust from the real place and the global constraint from finite places are **completely orthogonal** operations in mathematics:

- **Real place** $\rightarrow$ **commutator** $[D, a]$: local differential structure, generates dynamics, responsible for fluctuations
- **Finite places** $\rightarrow$ **trace** $\text{Tr}_\omega(\cdot)$: global topological invariant, generates rigidity, responsible for constraints

The two do not interfere: - Commutators change local structure but **do not change the value of the trace** (cyclicity of trace: $\text{Tr}(ab) = \text{Tr}(ba)$; trace of a commutator is zero) - The trace sees the global total but **does not care about local arrangement** (trace only sees the sum, not the order)

This is why they cooperate perfectly—one governs “how to push,” the other governs “what is the total number”—operating along orthogonal mathematical dimensions independently, without mutual information leakage, constituting a complete set of non-interfering coordinates.

1.9 9 Conclusion

The two-layer spectral triple constructed in this paper provides a complete noncommutative geometric model for the Hilbert-Plya realization of Riemann zeros. The core conclusions are summarized in six points:

1. **Local archimedean foundation:** Defining the Dirac operator $D_{\text{loc}} = 2i(\partial_{X_1} - \partial_{X_2})$ on $L^2(\mathbb{R}_+^\times \times \mathbb{R}_+^\times)$, its spectral determinant precisely yields the archimedean factor $\pi^{-s/2}\Gamma(s/2)$. The $-\frac{1}{2}$ shift in the Hopf algebra comultiplication locks the $\frac{1}{2}s(s-1)$ factor in the spectral parameter expansion, providing an intrinsic algebraic source for the functional equation symmetry.
2. **Global adelic lift:** Lifting the Hilbert space to the cuspidal automorphic function space $L_0^2(\text{GL}_2(\mathbb{Q}) \backslash \text{GL}_2(\mathbb{A}))$ and introducing contributions from all finite prime places through the restricted tensor product $D = D_\infty \hat{\otimes} \bigotimes_p' D_p$. The global spectral determinant is identically equal to the Riemann ξ function factor by factor, realizing a bidirectional one-to-one correspondence between operator spectrum and zero set.
3. **Dynamical origin of the critical line:** The critical line $\Re(s) = 1/2$ is a theorem, not a hypothesis. The self-adjointness constraint $\beta_1 = \beta_2$ and the functional equation pairing constraint $\rho_2 = 1 - \rho_1$ jointly force $\beta = 1 - \beta$, deriving the critical line from first principles without any a priori presupposition on zero locations.
4. **Zero-prime duality:** Through the pairing between spectral measure and Dixmier traces, a strict Fourier duality is established between zero imaginary part oscillations and prime-power discrete supports, fully reproducing the bilateral structure of the Weil explicit formula; the arithmetic trace formula and cyclic homology pairing are strictly equivalent.
5. **Mechanistic explanation of GUE universality:** The free independence of the crossed

product algebra guarantees that spectral statistics fall into the Wigner semicircle law/GUE universality class in the large-cutoff limit, explaining the GUE statistical properties of Riemann zero level spacings from the level of noncommutative structure.

6. **Three-layer structure of fluctuations and scale separation:** No fluctuation at the spectral determinant level (algebraic identity), no fluctuation at the trace formula level (distributional equality), and at the statistical level a scale separation between small-scale GUE fluctuations and large-scale analytic rigidity. Fluctuations originate solely from noncommutative dynamics over the real place; finite-place contributions are absorbed into the overall analytic structure, with the two operating along orthogonal dimensions of commutators and traces independently.

This construction depends on no unproved spectral conjecture. The local and global layers correspond respectively to the Γ -factor and Euler product of the Riemann $\xi(s)$, with their coupling naturally realized by the diagonal embedding of $\mathrm{GL}_2(\mathbb{Q})$. The critical line emerges as an inevitable corollary of self-adjoint operator spectral theory and algebraic module structure. The scale-separated image of fluctuations provides a unified mechanistic explanation for the coexistence of the Montgomery–Odlyzko law and zero analytic rigidity.

1.10 References

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